

ASWIN RAVICHANDRAN

(929) 595-3080 | aswin.r1809@aogjob.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

Data Engineer with over 4 years of experience in healthcare and business analytics, specializing in designing and implementing scalable ETL/ELT pipelines, data warehousing solutions, and predictive modeling. Skilled in SQL, Python, and big data technologies such as Apache Spark and Databricks. Experienced in leveraging cloud platforms (AWS, Azure, GCP) and tools like Snowflake and Airflow to optimize data workflows and drive business intelligence.

TECHNICAL SKILLS

Programming & Databases: Python (Pandas, NumPy, PySpark), SQL, Java, MySQL, PostgreSQL

Cloud Platforms: AWS (Glue, Redshift, S3, Lambda, EMR), Azure (Data Factory, Synapse, Databricks), GCP

Big Data Tools: Apache Spark, Hadoop, Kafka, Airflow, Snowflake, Databricks

Data Engineering: ETL/ELT pipelines, Data Warehousing, Dimensional Modeling (Star/Snowflake Schema), SCD

BI & Visualization: Power BI, Tableau, Azure Analysis Services

DevOps: Docker, Kubernetes, CI/CD (Jenkins), Git/GitHub

WORK EXPERIENCE

Mount Sinai Health System, NYC, USA

Mar 2023 - Present

Data Engineer

- Developed and maintained scalable ETL/ELT processes using AWS Glue, Apache Spark, and PySpark, ensuring efficient data ingestion and transformation.
- Extracted data from multiple sources, including AWS DynamoDB, external OLTP systems, and RESTful APIs.
- Extracted data from PostgreSQL, MySQL, and RESTful APIs, ensuring consistency and reliability.
- Constructed scalable data lakes on AWS S3 to consolidate structured and unstructured data for analytics.
- Configured Amazon Redshift as the central data warehouse, optimizing data loading processes and resolving ingestion errors using AWS CloudWatch.
- Set up real-time monitoring and alerting systems for data pipelines using tools like AWS CloudWatch.
- Developed a healthcare data pipeline to ingest FHIR-formatted data via HL7 APIs, improving data accessibility for over 200 clinicians.
- Designed data processing workflows integrating AWS services, ensuring optimal utilization of EC2 resources and efficient data transfer between S3 and EMR.
- Utilized Apache Spark for distributed data processing, handling datasets exceeding 100 million records.
- Created tables, stored procedures, and extracted data using T-SQL for business users to perform data analysis and generate technical solutions.
- Developed multiple ETL pipelines integrating data from diverse sources using Spark and Lambda, based on pre-set scripts.

Version 1 , IND

Jan 2020 – Jul 2022

Associate Data Engineer/Analyst

- Designed and implemented robust data pipelines on Azure, integrating Azure Data Factory for data ingestion from diverse on-premises and cloud sources.
- Utilized Azure Data Lake to store raw data in formats such as JSON files.
- Employed Python and SQL within Azure Data Factory to execute complex data transformations, cleaning, and enrichment processes.
- Utilized Apache Airflow for advanced workflow orchestration, ensuring reliable scheduling and dependency management across multiple ETL jobs.
- Implemented Delta Lake on Azure Databricks for ACID transactions on large datasets, enabling scalable analytics.
- Developed scalable ETL pipelines leveraging Azure Databricks for distributed data processing, using Python and PySpark APIs for efficient data transformations and loading into Azure Data Lake Store.
- Leveraged PySpark in Databricks for data transformations and cleaning, integrating SQL-based validations to maintain high data quality.
- Built a multi-terabyte Azure Synapse Analytics warehouse to handle large-scale ingestion, transformation of millions of records (ELT).
- Developed interactive Power BI dashboards for clinical teams, visualizing patient data trends and enabling data-driven decisions.
- Set up monitoring for Azure Data Factory pipelines using Azure Monitor, tracking data flow and triggering alerts for failed activities.
- Loaded processed data into Snowflake, optimizing query performance and enabling near real-time analytics.
- Utilized Snowflake for data warehousing, implementing optimized storage and retrieval solutions to support scalable analytics and reporting.

CERTIFICATION

[Azure Data Engineer Associate](#) – Microsoft

[Architecting with Google Compute Engine](#) - Coursera

EDUCATION

Master of Science in Computer Science, New Jersey Institute of Technology, NJ, USA

August 2022 – December 2023

Bachelor of Technology in Computer Science, Rajalakshmi Engineering College, TN, India

August 2016 – May 2020